DIRECTIONAL RESIDUAL STABILISATION FOR HIGH-PÉCLET-NUMBER ADVECTION–DIFFUSION PROBLEMS: GPU-ACCELERATED IMPLEMENTATION IN THE VANADIS SYSTEM

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1. INTRODUCTION

Atmospheric pollutant transport frequently involves advection-dominated regimes in which a conventional finite-element Galerkin discretisation can develop non-physical spatial oscillations. The difficulty becomes more pronounced on strongly non-uniform three-dimensional meshes, where the element dimension encountered by the flow changes substantially across the domain.

This paper presents the current Directional Residual (DR) formulation implemented in Vanadis, a three-dimensional transient Eulerian finite-element model for atmospheric pollutant transport. The numerical core of Vanadis originated in earlier finite-element work on atmospheric dispersion by Chodorski and Pietrzyk (1997) and Chodorski and Malinowski (1999). The current implementation combines flow-directed stabilisation with an implicit transient formulation, eight-node hexahedral elements, OpenMP-parallel element processing, and an Element-by-Element (EbE) GPU solution strategy.

The purpose of the DR formulation is to stabilise the transport terms in advection-dominated conditions while leaving the prescribed physical diffusion operator in its standard Galerkin form. The stabilisation strength is evaluated locally from a streamwise Péclet number so that it follows both the local mesh geometry and the direction of transport.

2. GOVERNING EQUATION AND FEM FORMULATION

Vanadis solves the transient advection–diffusion–reaction equation

$$\frac{\partial S}{\partial t} + v_x\frac{\partial S}{\partial x} + v_y\frac{\partial S}{\partial y} + v_z\frac{\partial S}{\partial z} - \frac{\partial}{\partial x}\left(K_x\frac{\partial S}{\partial x}\right) - \frac{\partial}{\partial y}\left(K_y\frac{\partial S}{\partial y}\right) - \frac{\partial}{\partial z}\left(K_z\frac{\partial S}{\partial z}\right) + PS = Q_v. \quad (1)$$

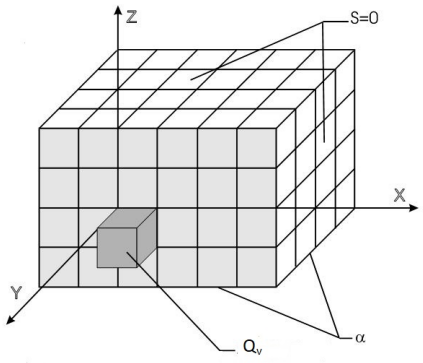


Fig. 1. Schematic representation of the Vanadis three-dimensional finite-element domain. Q_v denotes the volumetric source, α the surface-exchange/deposition coefficient, and $S = 0$ the homogeneous external Dirichlet boundary condition.

where S is pollutant concentration, v_x, v_y, and v_z are the velocity components, K_x, K_y, and K_z are physical diffusion coefficients, P is a first-order removal coefficient, and Q_v is a volumetric source.

The domain is discretised using trilinear eight-node hexahedral (HEX8) finite elements with $2\times2\times2$ Gaussian integration. A volumetric source can be assigned to individual elements. A Robin boundary condition is used for surface exchange/deposition, while homogeneous Dirichlet conditions are applied at the other external boundaries. Surface integration is performed on the physical three-dimensional HEX8 face, which allows inclined and moderately distorted boundary surfaces. The boundary configuration used in the present Vanadis setup is illustrated in Fig. 1.

3. DIRECTIONAL RESIDUAL STABILISATION

For advection-dominated transport, the standard Galerkin test function is replaced by the directional

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$$\boxed{W_i = N_i + \tau_{\mathrm{DR}} \left(v_x \frac{\partial N_i}{\partial x} + v_y \frac{\partial N_i}{\partial y} + v_z \frac{\partial N_i}{\partial z} \right)}. \tag{2}$$

Here N_i is the standard HEX8 interpolation function and τ_{DR} is the local stabilisation parameter. The formulation is related to the streamline upwind/Petrov–Galerkin approach of Brooks and Hughes (1982), while the Vanadis implementation uses its own coupling of directional weighting, transient integration, boundary treatment, and EbE CPU/GPU solution.

The same directional weighting is used in the transient mass term and the advective contribution. The transient matrix is therefore

$$C_{\mathrm{DR},ij} = \int_{V_e} W_i N_j \, dV. \tag{3}$$

For physical diffusion, the conventional Galerkin gradients of the interpolation functions are retained. Thus DR changes the transport weighting without modifying the prescribed physical diffusion coefficients.

The element equations can be written compactly as

$$[C_{\mathrm{DR}}]\{\dot{S}\} + [H]\{S\} = \{L\}, \tag{4}$$

where $[H]$ contains the diffusion, advection, first-order removal, and surface-exchange contributions, while $\{L\}$ contains the volumetric and boundary source terms.

The parameter τ_{DR} is evaluated locally from a streamwise Péclet number. The characteristic element dimension is determined in the actual flow direction from the local element geometry, and the effective physical diffusivity is evaluated in the same direction. Consequently, the stabilisation responds automatically to mesh stretching, element orientation, and directional diffusion. Numerically stable limiting evaluations are used for very small and very large Péclet numbers. These implementation details are inexpensive and are evaluated locally at the integration points.

4. STEADY AND TRANSIENT SOLUTION

For a steady calculation, Eq. (4) reduces to the linear system

$$[H]\{S\} = \{L\}. \tag{5}$$

The current transient implementation uses the generalized trapezoidal scheme with $\theta = 2/3$. In the notation used by Vanadis, the matrix and right-hand side for one time step can be written as

$$\begin{aligned} [H^*] &= 2[H] + \frac{3}{\Delta t}[C_{\mathrm{DR}}], \\ \{L^*\} &= -\left([H] - \frac{3}{\Delta t}[C_{\mathrm{DR}}] \right) \{S^n\} + 3\{L\}, \\ [H^*]\,\{S^{n+1}\} &= \{L^*\}. \end{aligned} \tag{6}$$

The method is implicit, one-step and A-stable. It has the same stationary limit as Eq. (5), but its transient damping differs from the BDF2 formulation described by Gear (1971). A separate comparison with BDF2 was performed for the same DR spatial operator and $\Delta t = 100$ s. Both integrations converged to essentially the same final steady-state mass. In this test, the $\theta = 2/3$ solution showed a smoother transient and required 1247 DBCG iterations in total, compared with 1316 for BDF2 (about 5.5% more). This comparison is included only to characterise the tested transient behaviour and is not presented as a general accuracy ranking of the two time-integration methods.

5. REPRESENTATIVE HIGH-PÉCLET TEST

A representative transient calculation uses a 1.0 kg s^{-1} volumetric source, isotropic physical diffusion $K_x = K_y = K_z = 40$ m^2 s^{-1}, a first-order removal coefficient $P = 7 \times 10^{-4}$ s^{-1}, velocity components $(v_x, v_y, v_z) = (0, -5, 0)$ m s^{-1}, and a surface-exchange coefficient $\alpha = 0.01$ m s^{-1}. The mesh contains 2166 HEX8 elements and 2800 nodes; the time step is 100 s and the transient is followed to 2000 s. The principal parameters are summarised in Table 1.

The mesh is strongly non-uniform in the flow direction, with local spacing from 50 to 2500 m. For this axis-aligned test case, the current streamwise definition used by DR gives local Péclet numbers from approximately 3.125 to 156.25. Thus a single calculation contains both moderately and strongly advection-dominated elements, and the stabilisation parameter adapts locally as the plume moves through regions of different mesh resolution.

6. EARLY-TIME MASS-BALANCE VERIFICATION

An additional transport calculation was used as a global mass-balance check with the first-order removal and surface-deposition terms disabled ($P = 0$ and $\alpha = 0$) and without imposed Dirichlet concentration constraints. The source remained $Q = 1$ kg s^{-1} and the velocity remained $(v_x, v_y, v_z) = (0, -5, 0)$ m s^{-1}, so the DR stabilisation was active. Before boundary losses become appreciable, the integrated

Table 1. Representative Vanadis transient test.

Parameter	Value
HEX8 elements	2166
Nodes	2800
Source Q	1.0 kg s^{-1}
$K_x = K_y = K_z$	40 m^2 s^{-1}
$\lvert v_y \rvert$	5 m s^{-1}
First-order coefficient P	7×10^{-4} s^{-1}
Surface coefficient α	0.01 m s^{-1}
Time step	100 s
Final transient time	2000 s
Local streamwise Pe	3.125–156.25

Table 2. Early-time mass-balance check for $Q = 1$ kg s^{-1}, $P = 0$ and $\alpha = 0$.

Time [s]	Qt [kg]	Vanadis [kg]	Difference [%]
100	100	100.170	+0.17
200	200	201.618	+0.81
300	300	306.234	+2.08
400	400	413.063	+3.27
500	500	515.130	+3.03
600	600	603.073	+0.51
700	700	671.630	-4.05
800	800	721.118	-9.86

pollutant mass is expected to follow the cumulative source input, $M_{\mathrm{in}} = Qt$.

The mass is evaluated independently after every time step by three-dimensional $2 \times 2 \times 2$ Gauss integration of the HEX8 concentration field. As shown in Table 2, through $t = 600$ s the integrated mass remains within approximately 3.3% of the cumulative source input. At later times the integrated mass falls below Qt as the advected and diffused plume reaches the finite-domain boundary and pollutant leaves the computational domain.

7. ATMOSPHERIC DISPERSION COMPARISON

An atmospheric dispersion case was compared with the Pasquill-type calculation previously used for verification of the three-dimensional FEM model by Chodorski and Malinowski (1999). The horizontal and vertical concentration fields are compared in Fig. 2.

The two calculations show similar plume position, overall geometry, lateral spread, vertical spread, and concentration scale. In the horizontal comparison, both calculations reach concentration levels of approximately 2.2×10^{-2} mg m^{-3}. In the vertical cross-section, the highest plotted levels are approximately 9.0×10^{-3} mg m^{-3} for the Pasquill calculation and 8.5×10^{-3} mg m^{-3} for Vanadis.

The comparison is not intended to imply pointwise equivalence between a three-dimensional Eulerian FEM solution and a Pasquill dispersion formulation. Vanadis solves the transport equation on a finite computational domain with explicit boundary conditions, so the outer plume can be influenced by the zero-concentration lateral and upper boundaries if they are placed too close to the region of interest. Nevertheless, the comparison provides an independent physical-scale check of plume position, dimensions, and concentration magnitude.

8. CPU/GPU IMPLEMENTATION AND PERFORMANCE

The linear system is treated using an Element-by-Element representation. Local 8×8 HEX8 matrices are stored element by element and used directly in the iterative matrix-vector operations, so a conventional assembled global sparse matrix is not required by the solver. This has a substantial memory advantage for large three-dimensional meshes. For the approximately 2.8-million-node benchmark, the EbE matrix data occupy about 1.4 GB. For comparison, storage estimates for the same problem are approximately 66 TB for a full dense matrix and 333 GB for the banded representation used in the comparison. The three storage estimates are summarised in Table 3.

Table 3. Approximate matrix-storage requirements for the 2.8-million-node benchmark.

Matrix representation	Storage
Full dense matrix	$\sim$66 TB
Banded matrix	$\sim$333 GB
Vanadis EbE matrices	$\sim$1.4 GB

The nonsymmetric system is solved with a diagonally preconditioned BiConjugate Gradient method derived from the SLATEC/SLAP DBCG implementation of Greenbaum and Seager (1989). A CUDA version performs the iterative solution stage on the GPU using the same element-level numerical operator as the CPU path.

Element matrix and vector construction can additionally be parallelised on the CPU using OpenMP. Thread-local element workspaces and OpenMP reductions are used so that element calculations remain independent while the global vector and diagonal contributions are combined safely. The mea-

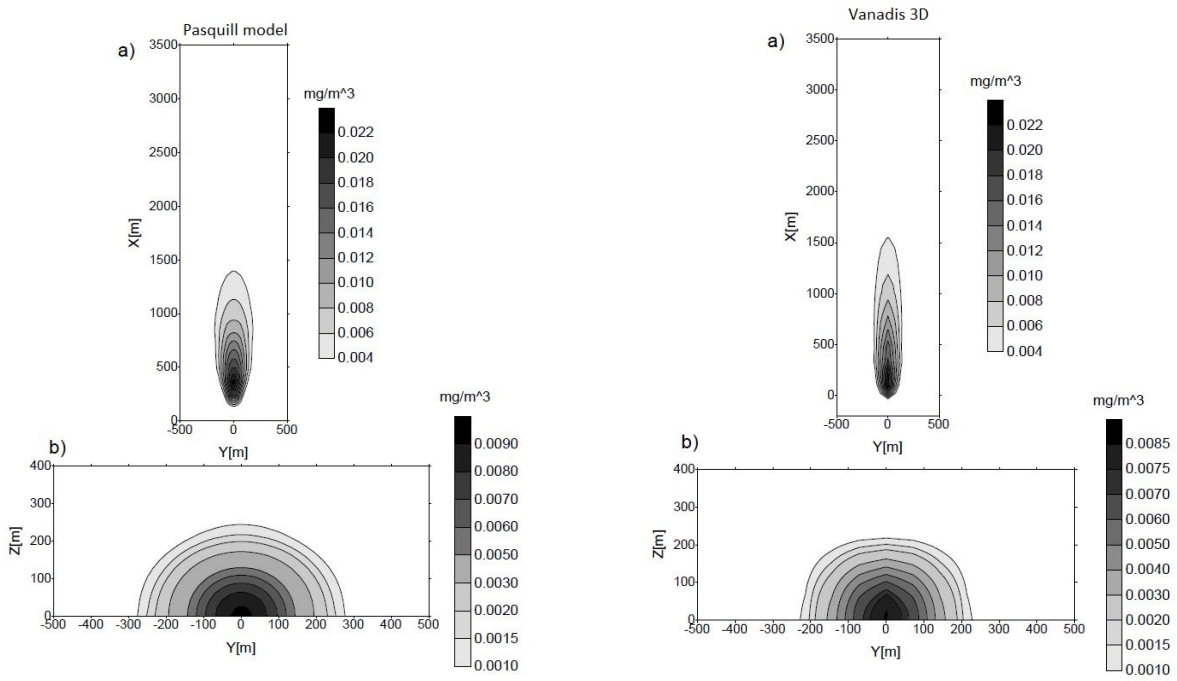


Fig. 2. Comparison of the Pasquill dispersion model (left) and Vanadis 3D (right). In each plot, panel (a) shows the horizontal plume distribution and panel (b) the vertical cross-section. Concentrations are in mg m^{-3}.

Table 4. OpenMP element-assembly performance for the 2166-element test.

Threads	Assembly time [s]	Speed-up
1	0.222	1.00
2	0.121	1.83
3	0.087	2.55
4	0.071	3.13

sured assembly scaling is given in Table 4.

The one-thread and four-thread calculations reproduce the integrated mass at 2000 s to a relative difference of approximately 2.4×10^{-12}. The remaining difference is consistent with the changed floating-point summation order produced by parallel reduction rather than a change in the numerical solution.

A separate large-mesh benchmark used approximately 2.8 million computational nodes and 20 time steps over 0–2000 s. The complete simulation required approximately 20 min on an Intel i5-4590 CPU and approximately 10 min when the iterative solution stage was executed on a GeForce GTX 1060 GPU. During the transient, representative DBCG solve times decreased from about 20.8 to 9.6 s per step on the CPU and from 8.9 to 6.6 s on the GPU as the iteration count decreased. These results were obtained on comparatively modest consumer hardware.

CPU and GPU calculations reproduce the same physical solution to floating-point precision, with the same convergence behaviour to the level relevant for the reported simulations. For multi-million-node cases, formatted output and post-processing I/O can account for a substantial part of the total runtime, so solver time and complete simulation time should be distinguished.

The complete Vanadis source code used for the calculations reported here is publicly available in the GitHub repository.

9. DISCUSSION

The current Vanadis architecture combines directional Petrov–Galerkin weighting, an implicit transient formulation, surface exchange/deposition, and a low-memory EbE iterative solution strategy suitable for GPU execution. The principal numerical point is that DR changes the flow-directed test weighting while the physical diffusion terms retain their conventional finite-element form.

The stabilisation is local and geometry-aware: its streamwise Péclet number is evaluated using the actual transport direction, the local element geometry, and the physical diffusivity in that direction. This is useful for environmental calculations with

stretched meshes, inclined surfaces, and locally distorted three-dimensional elements, but the detailed geometric evaluation remains a local implementation operation rather than an additional global model equation.

The present work does not claim that directional weighting constitutes a fundamentally new class of stabilisation distinct from established streamline Petrov–Galerkin methods. The contribution is the particular coupling and computational realisation in Vanadis: DR-weighted transient and advective terms, unchanged physical diffusion gradients, implicit time integration, Robin surface exchange, OpenMP element processing, and an EbE CPU/GPU solver.

10. CONCLUSIONS

Vanadis provides a three-dimensional finite-element framework for transient atmospheric advection–diffusion–reaction problems extending into strongly advection-dominated regimes. In the representative non-uniform test considered here, the local streamwise Péclet number ranges from approximately 3.1 to 156.3 within a single mesh.

The DR formulation applies directional weighting to the transient and advective terms while retaining the physical diffusion operator in conventional Galerkin form. The current $\theta = 2/3$ time integration is implicit and A-stable. In a separate mass-balance check with $P = 0$ and $\alpha = 0$, the integrated mass remains within approximately 3.3% of the cumulative source input through 600 s before boundary losses become appreciable.

Comparison with a Pasquill dispersion calculation gives similar plume location, dimensions, and concentration scale. CPU/OpenMP tests show approximately 3.13$\times$ acceleration of element assembly on four physical cores while maintaining numerical agreement to about 10^{-12} in relative integrated mass. For the 2.8-million-node benchmark, the EbE matrix data require about 1.4 GB, compared with approximately 66 TB for a full dense matrix and 333 GB for the banded comparison. The same benchmark demonstrates that the EbE/CUDA solution can be executed effectively on consumer-class GPU hardware.

The complete implementation is publicly available, allowing the numerical formulation, boundary treatment, CPU/GPU pathways, and reported calculations to be independently inspected and reproduced.
Project page: `https://marek-ac.meri.pl/`
Source code: `https://github.com/marek-ac/vanadis_solver-directional_residual`
Archived release: `https://doi.org/10.5281/zenodo.21352014`

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AI-assisted tools were used to support selected code refactoring and programming automation, as well as language editing of the manuscript.

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